

Figure 4: Estimation of the exponential of value function at $t = 0.5$ for a 1000-dimensional system by path integral MC (left), and by the proposed method (right).

$$\mu(\xi, T) = \exp\left(-\frac{1}{500}\xi_1^2 - \frac{1}{500}\xi_2^2\right). \quad (46)$$

With that, we generate data for $\bar{\varphi}(x, t)$ on spatial-temporal space $\Omega \times \mathbb{T} = [1, 2]^2 \times [0, 1.5]$, with grid size $d\xi = 0.1$ and $dt = 0.1$ and train the PINN on $\Omega \times \mathbb{T} = [1, 2]^2 \times [1, 1.5]$. We use a PINN with 3 hidden layers and 32 neurons per layer to learn the value function $\hat{\varphi}$. The activation function is chosen as hyperbolic tangent function ($\tanh$). We use the Adam optimizer (?) for training with initial learning rate set as 0.001. The PINN parameters θ is initialized via Glorot uniform initialization and the weights in the loss function (22) are set to be $\omega_p = \omega_d = 1$. The simulation is constructed based on the DeepXDE framework (?). Fig. 4 shows the estimated value function from the path integral MC and the proposed method. It can be seen that the proposed framework is able to estimate value functions accurately, while the path integral method has significantly more noise. Also, the computation time for training the PINN is significantly less than sampling path integral MC (80s v.s. ~ 3000 s). Note that the PINN is able to estimate the value function at unseen regions in the state space and generalize to longer time horizons, as the testing data at $t = 0.5$ is not seen by the PINN during training. We refer readers to the appendix for the safety probability estimation setting where similar results can be obtained.

Sample Complexity

We further examine the computation complexity of the proposed method to show its advantages in sample efficiency. We consider the problem of value function estimation of a 3-dimensional system with similar dynamics and cost defined in (40) and (41). Please see appendix for details of the settings. Fig. 5 shows the percentage error of the estimated value function with different number of samples for path integral MC and PINN with and without using comparison theorem for dimension reduction. The path integral MC with dimension reduction has sample complexities that are one order less than MC without dimension reduction, which indicates the efficacy of the dimension reduction scheme from the comparison theorem. The PINN further reduces sample complexity and achieves the best trade-off between accuracy and computation.

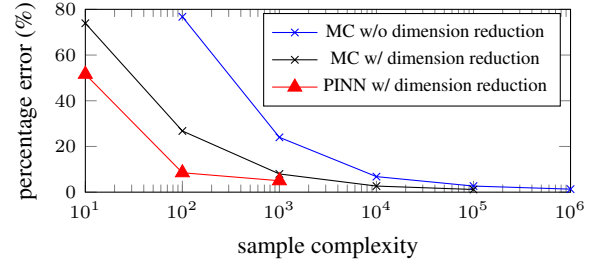


Figure 5: Percentage error of the estimated value function with path integral MC.

Feature Learning

We verify our autoencoder-like network by learning the features and cost function in equations (42) and (44) for the 3-dimensional system described above. The network consists of 5 fully connected hidden layers of sizes 100, 10, 2, 10, 100, respectively, with the activation function as hyperbolic tangent. We use the Adam optimizer with initial learning rate set as 0.001. The parameters of the network are initialized via the Glorot uniform initialization. The weights in the loss function (37) are set as $w_{RC} = 1$, $w_{C.T.} = 10$. The network is trained on a $[0, 1]^3$ state-space with grid size 0.01. The network successfully identified the features derived analytically in (??) with $\text{MSE}(\xi_1, \hat{\xi}_1) = 0.2$ and $\text{MSE}(\xi_2, \hat{\xi}_2) = 0.06$.

Conclusion

We propose a unified framework for value function and safety probability estimation of high-dimensional stochastic systems. The novel dimensionality reduction technique uses the comparison theorem to generate low-dimensional stochastic processes that provide an exact characterization of the cost/barrier function, significantly improving sample complexity. We then transform the low-dimensional process into a low-dimensional PDE, and leverage physics-informed learning to generalize solutions into longer time horizons and unseen regions of state-space. We also achieve automatic feature identification through a specially designed autoencoder-like neural network. Experiment results show the efficacy of the proposed method. Future work includes application to multi-agent robotic control systems.

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